Bacteriocins and the evolution of bacterial symbioses

Matishalin Patel, Stuart West

1 Abstract

Cooperative symbionts enable their hosts to exploit a diversity of environments. A low genetic diversity (high relatedness) between the symbionts within a host is thought to favour cooperation by reducing conflict within the host. However, hosts will not be favoured to transmit their symbionts in costly ways that increase relatedness, unless this also provides an immediate fitness benefit to the host. We suggest that antimicrobial warfare, with compounds such as bacteriocins, could provide a relatively universal reason for why hosts would gain a benefit from increasing the relatedness between bacterial symbionts. We theoretically test this hypothesis with a simple illustrative model that examines whether hosts should manipulate relatedness, and an individual based simulation, where hosts and symbionts can coevolve. We find that hosts can be favoured to manipulate relatedness, to reduce conflict between symbionts, and that this can in turn, select for more cooperative symbionts

keywords:

virulence, cooperation, game theory, kin-selection, evolution, social evolution

2 Introduction

Symbiotic cooperation between species allows new niches to be exploited, and the evolution of more complex life (Bourke, 2011; Douglas, 2014). Aphids can grow feeding only on nutrient poor plant sap, because their symbiotic bacteria provide essential amino acids (Baumann et al., 1995). The siboglinid worms depend upon symbiotic methane or sulphides eating bacteria to survive near hydrothermal vents (Thornhill et al., 2008). Symbiosis between an archeal host cell and an aerobic bacterium gave rise to the eukaryotes.

Both theory and empirical data have suggested that a high genetic relatedness between the symbionts within a host can play a key role in promoting symbiotic cooperation. When symbionts are more closely related, any benefits received from increasing host growth are more likely to be shared with relatives, which increases the kin selected benefit of providing help to hosts (Frank, 1994a,b; West et al., 2002b; Foster and Wenseleers, 2006; Leeks et al., 2019). Consistent with this prediction, across different symbioses, bacterial symbionts appear to provide more help to their host when relatedness is higher (Fisher et al., 2017). Additionally, experimental and observational studies have shown that conditions which lead to a higher relatedness also lead to cooperation (Sachs and Wilcox, 2006; Dubilier et al., 2008; Bennett and Moran, 2015).

However, Frank (1996) pointed out that although a higher relatedness would favour symbionts to become more cooperative, hosts would not necessarily be selected to transmit or house their symbionts in ways that increased relatedness. The reason for this is that, assuming cooperation was not adjusted conditionally in response to the local relatedness, symbionts would only evolve a higher level of cooperation in response to relatedness over time. Consequently, all else being equal, there would be no immediate fitness benefit to transmitting symbionts in a way that increased related-

ness.

We suggest that antimicrobial warfare through compounds such as bacteriocins could provide a relatively general explanation for why hosts would be selected to increase the relatedness between bacterial symbionts. Recent empirical work has shown that bacteria conditionally up-regulate their production of bacteriocins in response to the presence of competing clones (Majeed et al., 2011; Mavridou et al., 2018; Bhattacharya et al., 2018). Consequently, when there are more competing lineages and so relatedness between interacting bacterial symbionts is lower, bacterial symbionts will put more resources into killing each other This increased toxin production will reduce bacterial growth, such that the bacteria will be less beneficial to their hosts. We suggest that hosts will be selected to transmit or house their symbionts in ways that increased relatedness, to decrease this costly conflict. We theoretically model the plausibility of this hypothesis, and determine the conditions under which it would most be favoured.

We first develop a deliberately simple model, which captures our hypothesis in an easily interpretable way, which could be applied to many different cooperative mutualisms. We then perform a set of simulations of some simple scenarios to test our model with less restrictive assumptions.

3 Models

3.1 Illustrative Model

We assume that symbiotic bacteria inhabit a host. Empirical data has shown that when bacteria encounter competing strains, they increase their production of bacteriocins (Majeed et al., 2011; Mavridou et al., 2018; Bhattacharya et al., 2018). Therefore, when more bacterial strains infect a host, and so the genetic relatedness between symbionts is lower, the growth rate of each strain will be reduced for two reasons: (1) they will be investing more re-

sources into bacteriocin production; (2) the mortality caused by bacteriocins from other strains will be greater. We capture these effects by assuming that symbiont growth is a positive function of the relatedness within the group:

$$S(r) \propto r^{\alpha}, \quad \alpha \in \mathbb{R}_{>0}. \tag{1}$$

The parameter α determines whether the relationship is linear ($\alpha = 1$), decelerating ($\alpha < 1$) or accelerating ($\alpha > 1$). This shape parameter captures all the details of how the production and effect of the toxin interacts to influence the growth rate of a symbiont.

We assume that hosts gain more benefit from their bacterial symbionts when these symbionts are better able to grow. We ask whether a host would be favoured to invest in a costly manipulation to influence relatedness amongst their symbionts. This manipulative trait ($g \in [0,1]$) could be a behaviour, such as young eating parental faeces or a structural adaptation, such as specialised organs in the host to store and transmit symbionts (Douglas 1989).

We assume the initial relatedness amongst symbionts infecting a host is $r_s \in [0,1]$. This starting relatedness could be due to existing behaviours, structural constraints or transmission routes. We assume that if a host invests an amount $g \in [0,1]$ in a costly manipulation trait that this increases relatedness to some final relatedness ($r_f \in [0,1]$):

$$r_f = r_s + g(1 - r_s). \tag{2}$$

We assume that the cost of investing in the trait that increases symbiont relatedness reduces host fitness by a fraction $(1-g)^{\beta}$, where β is another shape parameter. We assume that the cost of increasing symbiont relatedness and benefits from symbiont growth interact multiplicatively, such that

host fitness (W_H) is:

$$W_H = (1-g)^{\beta}\,(r_s + g(1-r_s))^{\alpha} \tag{3}$$

We aim to find when this costly trait is favoured. To do this we solve for a candidate evolutionarily stable strategy (ESS) amount of manipulation for the host to invest in (g^*). The ESS (g^*) represents the strategy which could not be beaten by any other strategy, and is given by solving for when fitness is maximised; $\frac{dW_H}{dg}\Big|_{g=g^*} = 0$ (Smith and Price, 1973):

$$g^* = 1 - \frac{b}{(a+b)(1-r_s)}. \tag{4}$$

Equation (4) predicts hosts can be favoured to perform a costly trait that increases relatedness between symbionts, to reduce the extent to which symbionts invest in costly conflict. Specifically, the ESS host manipulation of relatedness (g^*) will increase when: (i) the starting relatedness amongst symbionts decreases (r_s), (ii) the influence of relatedness on symbiont growth is more accelerating (higher α), and (iii) the cost of manipulating relatedness is more decelerating (lower β).

3.2 Simulation

Our analytical model made a number of simplifying assumptions. In particular, we: (1) assumed a simple relationship between symbiont growth and host fitness; (2) did not specifically model symbiont cooperation, or examine how this would respond to host manipulation; (3) did not allow for co-evolution between host and symbiont traits. We explored the consequences of relaxing these assumptions with an individual-based simulation.

We performed two simulations. In the first simulation the symbionts unconditionally express bacteriocins into the environment, however these toxins did not target self. This means that when a host increases relatedness

between bacteria the toxins they produce do not affect each other and the group has a higher average growth rate.

This unconditional response is a simplification, though some species may pursue this strategy as a response to high levels of competition (Bhattacharya et al., 2018). In the second set of simulations we add in a conditional response on the part of the bacteria so that bacteria invest less in bacteriocin when they interact with member's of their own strain. This means relatedness has an effect similar to the first model in that bacteriocin cannot kill other individuals but also there is an extra benefit to the symbiont in that it can dedicate more energy to growth. This model also includes a host quality measure which decreases the more bacteriocin is produced. This models how bacteriocin production might harm the symbionts vertical transmission through reduced host quality.

3.2.1 Simulation with unconditional responses

We assume a population of N hosts in a birth-death Moran process (Zukewich et al., 2013). Each generation there are N individual reproduction events where one member of the host population copies itself, with a probability proportional to its fitness, and another member dies at random.

Each time a host is born, it acquires k symbiont lineages (strains or clones), which can be acquired either vertically from their parent, or horizontally, from the environment. We assume that the host (i) has an evolvable trait m_i that manipulates the likelihood of vertical transmission from its parent, $m \in [0,1]$. Specifically, we assume that the final probability of vertical transmission is given by:

$$\lambda_{f,i} = \lambda_s + m_i(1 - \lambda_s), \tag{5}$$

where, λ_s is the base chance of vertical transmission if the host does not attempt to manipulate it, and $\lambda_{f,i}$ is the final value. This probability is

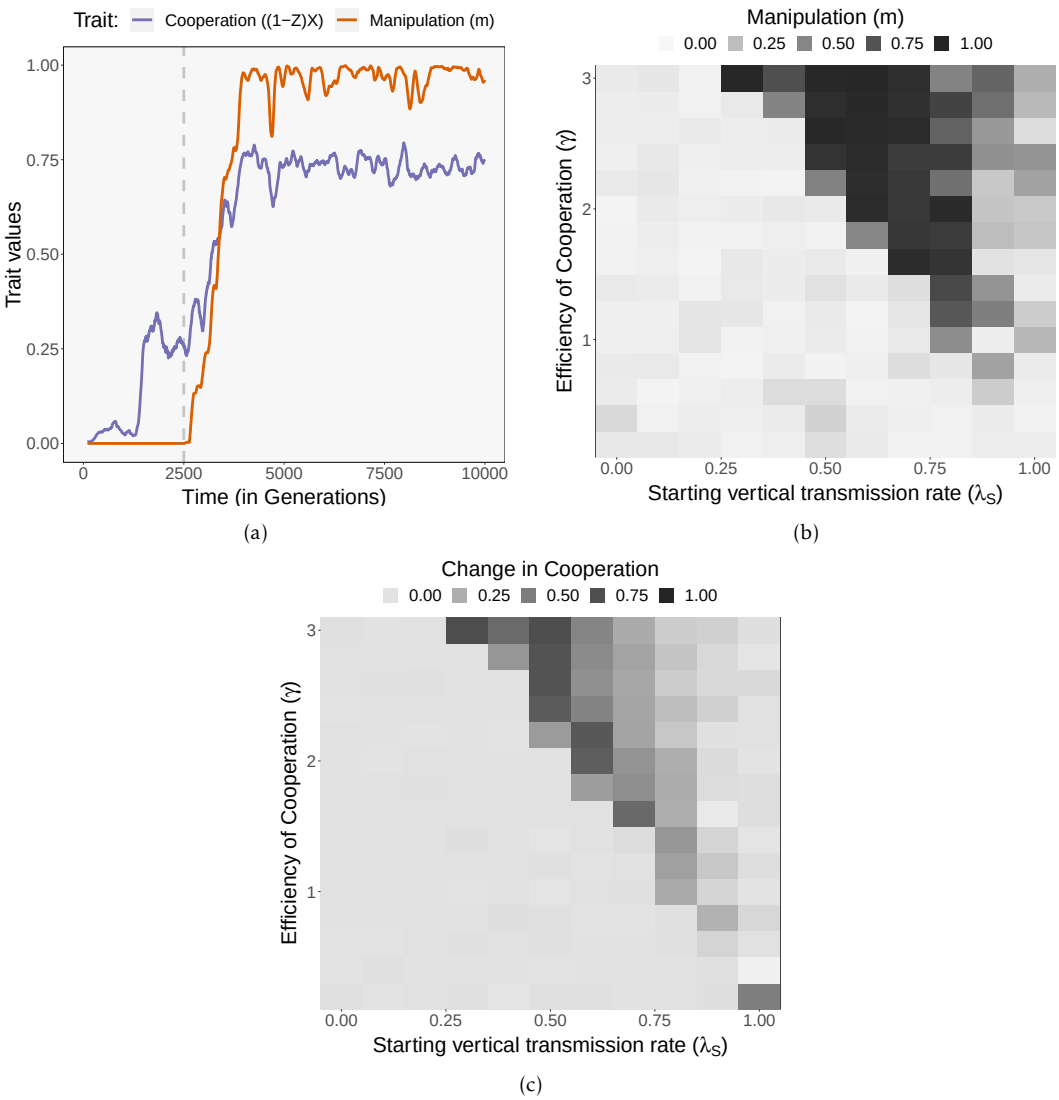


Figure 1: **(a)** An example simulation run with a starting vertical transmission rate of $\lambda_S = 0.5$ and an efficiency of cooperation $a = 2.4$. The gray dashed line indicates the time at which a host can invest in manipulation (orange). A mutant spreads and is followed by an increase in symbiont cooperation (purple). **(b)** Manipulation evolves in the host when starting vertical transmission is intermediate and when efficiency of cooperation is high. When starting relatedness is maximal there is no selection for investing in manipulation. At lower relatednesses manipulation is favoured as long as the benefit from cooperation is great enough. **(c)** Here we measure the change in symbiont cooperation from the symbiont evolved with no host manipulation and the symbiont after manipulation has evolved. In the lower boundary where evolution of manipulation evolves but starting vertical transmission and/or efficiency of cooperation are low is where manipulation has the largest impact on cooperation.

unique to each host based on its personal manipulation trait m_i. After a period of 2500 generations of burn-in the host trait for manipulation m is allowed to evolve from a starting value of 0.

So, with probability $\lambda_{f,i}$ when a host reproduces k symbionts are chosen, with a probability proportional to growth rate relative to the group, from the symbionts within it to colonise its child (vertical transmission). However, with probability $1-\lambda_{f,i}$ those k symbionts are instead chosen from the entire pool of all symbionts within all hosts, with a probability proportional to growth rate relative to the population (horizontal transmission).

We assume that symbionts are under two zero-sum tradeoffs. These are controlled by two evolving traits one for growth (Z) and one for cooperation (X). They may either invest in growth and replication ($Z \in [0,1]$) or the production of extracellular effectors ($1-Z$). Of these extracellular effectors they may either produce a beneficial function for the host ($X \in [0,1]$) or toxins that harm other bacteria ($1-X$). We allow the mutation, and therefore evolution, of Z, X within the symbionts using small deviations drawn from a normal distribution (Norm$(0, 0.1)$).

The relative growth rate of a symbiont will be determined by the amount it invests into growth, and the extent to which it is killed by the bacteriocins of other lineages, relative to the lineages that it is competing with. We assume that toxin's are discriminatory and do not target self we also assume that expression is not conditional (Bhattacharya et al., 2018). This unconditional expression still allows greater growth and less conflict at high relatedness because the toxin's do not target clones. L is the set of all strains (lineages) within a host, within each strain, $l \in L$, there is a set of individuals J. The growth rate ($S_{l',j'}$) for a focal symbiont $j' \in J$ of strain $l' \in L$ inside host i is given by:

$$S_{l'j'} = Z_{l'j'}\left(1 - \frac{1}{k-1}\sum_{l \in L \setminus l'}\sum_{j \in J}\left(1 - X_{l,j}\right)\left(1 - Z_{l,j}\right)\right) \tag{6}$$

This is the amount of energy put into growth by the symbiont ($Z_{l'j'}$) multiplied by the bracketed term indicating the effect of the toxin produced by others. The second term sums over each strain in the host, ignoring individuals of the same strain ($l \in L \setminus l'$), and each individual in that strain. It sums together the contributions from each individual towards toxin production ($(1 - X_{l,j})(1 - Z_{l,j})$). This entire quantity is then divided by the total number of symbionts (k) in the host excluding the focal individual ($1/(k-1)$) to give the average toxin per capita. This growth rate is calculated for each individual in the group and then divided by the group average to get the relative growth rate of each individual ($W_{S_{l'j'}}$):

$$W_{S_{l'j'}} = \frac{S_{l'j'}}{\overline{S}} \tag{7}$$

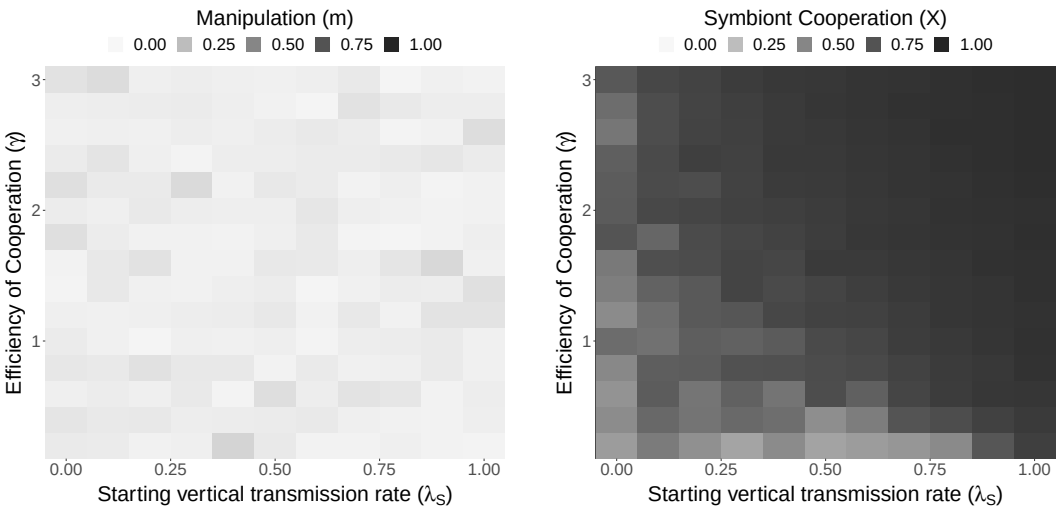
We assume that the fitness of a host (W_H) is given by the sum of the cooperative benefits given by each symbiont weighted by the symbiont relative growth ($W_{S_{l'j'}}$) rate to give the benefit to the host, multiplied by the proportion of resources not invested into symbiont manipulation:

$$W_{H_i} = \left(\sum_{lj} W_{S_{lj}} X_{lj} (1 - Z_{lj}) \right)^{\gamma} (1 - c\, m_i)\,, \tag{8}$$

where, c is the cost per unit of the manipulation trait g, and γ is the efficiency of cooperation.

We simulate populations of hosts using this framework for multiple different parameter sets. Figure 1a shows an example simulation run. After a period of burn-in (2500 generations) a mutant host invades that invests in manipulation. This increase in manipulation leads to a increase in symbiont cooperation ($(1 - Z)X$).

We found manipulation evolved when the starting vertical transmission rate is at an intermediate value. When it is too low the increased manipulation seems to not be effective enough and when it is too high symbionts



(a) (b)

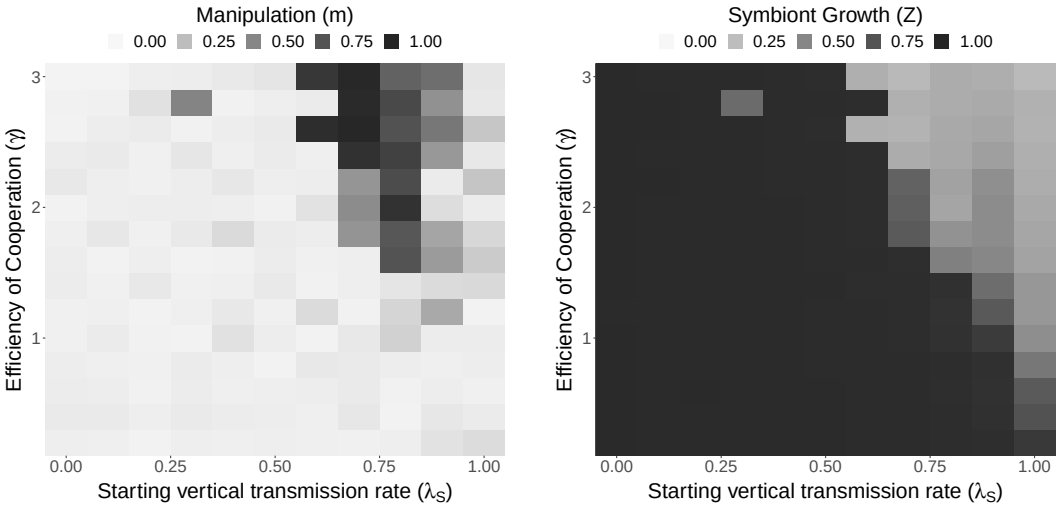
Figure 2: **(a)** When the evolution of growth is fixed, $Z = 0.5$, we find that host control doesn't evolve. **(b)** When holding growth at a constant ($Z = 0.5$) cooperation in the symbionts evolves to a high level despite low host control. This leads to no selection for the host to increase control.

are already highly cooperative so there is no benefit (fig. 1b). Figure 1c shows the difference between this basic level of cooperation and the level of cooperation at the end of a simulation once host manipulation has had an impact. Host manipulation changes symbiont cooperation the most in the same region as it evolves to the highest level.

Holding traits constant We considered four tests to confirm the simulations were acting as we expected. Firstly, when we hold the bacterial growth trait, $Z = 0.5$, constant and allow cooperation X to evolve we observe no evolution of host control (fig. 2). This occurs because holding growth constant leads to the evolution of high levels of cooperation amongst the symbionts ($X \sim 1$) as there is no benefit from bacteriocins when relatedness is low. This leads to no selection for the host to evolve control as there is low conflict regardless. Secondly, when we consider the case where cooperation, $X = 0.5$, is held constant this leads to limited evolution of host manipulation (fig. 3). When $X = 0.5$ symbionts optimise their growth rate Z to an intermediate value when the starting vertical transmission rate is high and when efficiency of cooperation is high (fig. 3b). Otherwise they evolve a growth rate

that is close to one ($Z \sim 1$). When the growth rate is intermediate this increases the production of bacteriocins ($(1 - Z)(1 - X)$). The host can then be selected for to increase relatedness and decrease the level of bacteriocins produced by the symbiont.

Thirdly we show that, when holding both cooperation and growth constant ($X = 0.5$, $Z = 0.5$), control cannot evolve. It may not immediately be clear why this should be the case as our illustrative model shows that manipulation should be stable in this scenario. One possible explanation is that evolutionary divergence between vertical and horizontal symbionts might be necessary to stabilise costly manipulation in our model. Consider a population of symbionts where each member is its own species (maximal conflict). Now a host population that is monomorphic for no manipulation would be invaded by a population that exerts a slight control (á la Taylor and Frank (1996)) as they would experience slightly less conflict. And this process should repeat until an equilibrium is reached, this logic is equivalent to the argument in section 3.1. However, our simulations use a finite population, by increasing relatedness within hosts there is also an increase in relatedness of the horizontally transmitting symbionts (transmission is probabilistic). This leads to the invasion of low manipulation hosts that take advantage of this low symbiont diversity. This exploitation means that manipulation cannot evolve when all symbionts are identical. The ability for symbionts to evolve with their hosts leads to low growth mutants that do not transmit horizontally as well. The feedback in the vertical lineages essentially separates the symbionts into vertical and horizontal pools and reduces the benefit from being a low manipulation mutant as horizontal transfer will be dominated by low quality symbionts.



(a) (b)

Figure 3: **(a)** When the evolution of cooperation is fixed, $X = 0.5$, we find that host control can evolve. **(b)** When holding cooperation at a constant ($X = 0.5$) selection for maximal growth ($Z = 1$) seem to relax at high vertical transmission rates (λ_s) and high efficiencies (γ). This selects for host mutants that reduce the conflict generated in those regions.

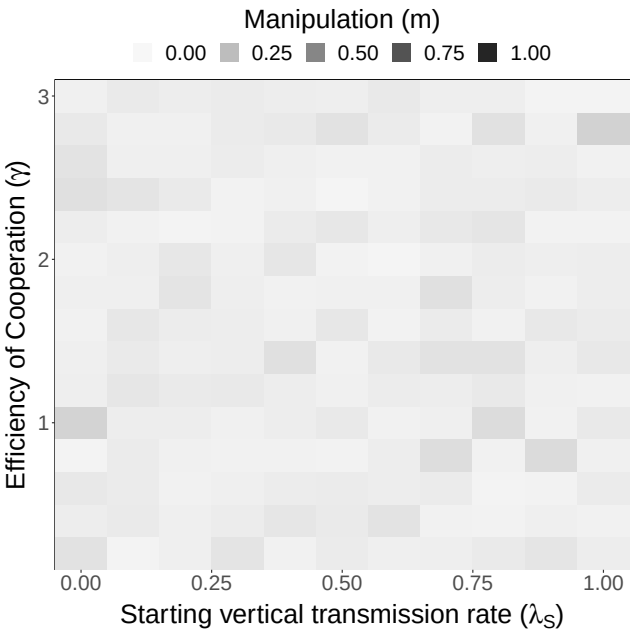


Figure 4: When growth and cooperation are fixed, $Z = 0.5$ and $X = 0.5$, host control does not evolve.

3.2.2 Simulation with conditional response

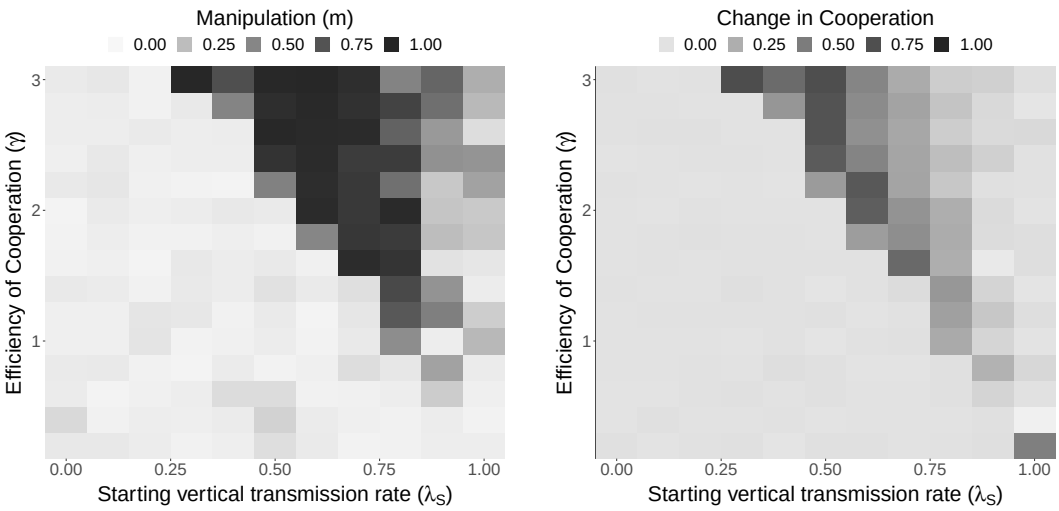
In the simulations performed in section 3.2.1 we had a discrimination system for the bacteriocin but it was not conditional to the presence of other strains. We also had a relative growth rate effect whereby the toxin produced in a patch only mattered for within host growth and everything was normalised between hosts this may have lead to the earlier simulation underestimating the effect of bacteriocin production when selecting bacteria for horizontal transmission.

To attempt to address this we made two changes to the equations presented in section 3.2.1. Firstly for $S_{l',j'}$ (eq. (6)), which is the growth rates of the symbiont within a host we modified the equation to include a term that describes the amount of toxin that would have been produced by the focal individual but was not. This represents the benefit to the symbiont of not expending energy on bacteriocin production when relatedness is high. This gives a modified symbiont growth rate $S^F_{l'j'}$:

$$\begin{aligned} S^F_{l'j'} =& Z_{l'j'}\left(1 - \frac{1}{k-1}\sum_{l\in L\setminus l'}\sum_{j\in J}\left(1-X_{l,j}\right)\left(1-Z_{l,j}\right)\right) \\ & \left(1 + \sum_{j\in J\setminus j'}\left(1-X_{l',j}\right)\left(1-Z_{l',j}\right)\right). \end{aligned} \tag{9}$$

The third term here is the sum of the toxin that would have been expended for each member of the focal individual's strain. This is then used to calculate a relative growth rate:

$$W^F_{S^F_{l'j'}} = \frac{S^F_{l'j'}}{\overline{S}} \tag{10}$$



(a) (b)

Figure 5: **(a)** Manipulation evolves in the host when starting vertical transmission is intermediate and when efficiency of cooperation is high. The patterns are similar to those found when doing the simulation without the explicit feedbacks (fig. 1b). **(b)** The change in symbiont cooperation from the symbiont evolved with no host manipulation and the symbiont after manipulation has evolved is shown here. In the lower boundary where evolution of manipulation evolves but starting vertical transmission and/or efficiency of cooperation are low is where manipulation has the largest impact on cooperation. This is again similar to the pattern observed before (fig. 1c).

The second change was made to the host fitness function, $W^F_{H_i}$:

$$W_{H_i} = \left(\sum_{lj} W^F_{S^F_{lj}} X_{lj} (1 - Z_{lj}) \right)^{\gamma} \frac{1}{k} \left(1 - \sum_{lj} \left(\sum_{l \in L \setminus l'} \sum_{j \in J} \left(1 - X_{l,j}\right)\left(1 - Z_{l,j}\right) \right) \right) (1 - c\, m_i) \,. \tag{11}$$

We insert a term after calculating the benefit from the cooperation produced by each symbiont, the new term calculates the amount of toxin produced by each symbiont within the host. This then penalises the cooperative benefit the symbionts given to the host. This captures the idea that host's with high levels of bacteriocin production will have a lower quality than those without as the symbionts will reach a lower carrying capacity within the host.

We also performed the same diagnostic simulations as those done for the first simulation with unconditional expression. We recovered the same patterns for our second simulation model (appendix A.1.1).

4 Discussion

We have shown that: (1) a host can be selected to increase relatedness amongst it's bacterial symbionts, to decrease conflict via antimicrobials such as bacteriocins; (2) this, in turn, selects for higher levels of cooperation from symbionts towards their hosts.

4.1 Illustrative model

Our illustrative model assumed that higher relatedness increases the growth rate of the symbiont. This is based upon experimental work on density dependent behaviour in bacterial warfare (Bashey et al., 2012; Mavridou et al., 2018). However, previous theory suggests that bacteriocin production may in fact follow a domed relationship with relatedness (Gardner et al., 2004). This would mean that increasing relatedness may decrease growth rates depending on the starting relatedness. This implies that in nature the evolution of host control may be constrained by the un-manipulated relatedness as sometimes selection may favour reducing relatedness. Indeed we might predict that evolution should favour either highly related symbionts or highly unrelated ones to avoid these intermediate relatedness values where conflict is high.

Our models predict that hosts will invest more resources into increasing relatedness when this will lead to a greater increase in fitness. For example, in systems where relatedness between symbionts would otherwise be very low, due to low rates of vertical transmission, or due to population structuring leads to greater mixing and conflict between different bacterial strains (fig. 1b). The diversity of mutualisms in the natural world would allow our predictions to be tested with across species comparative studies. For example, amongst the mycetocyte symbionts that infect a variety of insects the mode of vertical transmission varies from faecal smearing to specialised structures that transfer the symbionts from the mycetocytes to the ovaries

(Douglas, 1989). Previous empirical work on termites and ants has shown that they can be selected to maintain their fungal symbionts at high relatedness, to reduce incompatibility conflicts (Korb and Aanen, 2003; Poulsen and Boomsma, 2005; Aanen et al., 2009). Our hypothesis provides an analogous mechanism which could operate across all bacterial symbioses.

4.2 Simulations

The simulations agreed with the model in their patterns at the macro-scale. However the fact that holding the symbiont constant removed any selective pressure for the evolution of control shows the important dynamics that can be missed by static models. Our analytic model used a simple behavioural rule for the symbiont to show that costly manipulation could be favoured by the host with. Our simulations embedded a similar rule into a dynamic system with finite population sizes and we found that the evolution of manipulation in this scenario could not work when symbionts were all identical. We suggest that manipulation exerted by the hosts leads to a public goods dilemma were the good was the symbiont diversity. By investing in control hosts reduced the diversity for everyone including those who did not invest. Allowing the symbionts to evolve in response breaks this pattern. Evolution would lead to a separation between the symbiont lineages that are transmitted between hosts and within hosts. This separation breaks the advantage for low host manipulation mutants and stabilises manipulation in the host population. This argument for why evolution of control is not stable when symbionts cannot evolve could be tested properly by keeping track of the symbiont lineages during the simulation which was not a feature included in our simulation.

The simulations we used relied on abstraction when it came to most of the symbiont interactions. This was mainly done to spare computational time. Future work could improve on this by modelling the within host dy-

namics of the symbionts more explicitly. Modelling the individual symbiont interactions explicitly would also allow us to more easily keep track of the lineages of symbionts and confirm our suggestion that this bifurcation between strategies is what stabilises host manipulation.

4.3 Conclusion

We have focused on how a shared interest between symbionts and their hosts can favour symbiont cooperation. Analogous predictions could apply to cases where cooperation is enforced, via mechanisms such as sanctions or trading (Kiers et al., 2003, 2011). In those cases, enforcement works best when it is directed at clonal groups, such as within a legume nodule (West et al., 2002b,a; Wyatt et al., 2013). Reducing costly bacteriocin warfare provides a fitness benefit for hosts to do this, with the knock on influence that it would favour enforcement mechanisms, which would lead to even higher levels of cooperation. Consequently, reducing microbial warfare could facilitate symbiont cooperation more generally, irrespective of whether it arises via a shared interest or enforcement.

References

Duur K. Aanen, Henrik H. de Fine Licht, Alfons J. M. Debets, Niels A. G. Kerstes, Rolf F. Hoekstra, and Jacobus J. Boomsma. High Symbiont Relatedness Stabilizes Mutualistic Cooperation in Fungus-Growing Termites. *Science*, 326(5956):1103–1106, 2009. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.1173462.

F. Bashey, S. K. Young, H. Hawlena, and C. M. Lively. Spiteful interactions between sympatric natural isolates of Xenorhabdus bovienii benefit kin and reduce virulence: Spite benefits kin and reduces virulence. *Journal*

of Evolutionary Biology, 25(3):431–437, 2012. ISSN 1010061X. doi: 10.1111/j.1420-9101.2011.02441.x.

Paul Baumann, Linda Baumann, Chi-Yung Lai, Dadbeh Rouhbakhsh, Nancy A. Moran, and Marta A. Clark. GENETICS, PHYSIOLOGY, AND EVOLUTIONARY RELATIONSHIPS OF THE GENUS BUCHNERA: Intracellular Symbionts of Aphids. *Annual Review of Microbiology*, 49(1): 55–94, 1995. doi: 10.1146/annurev.mi.49.100195.000415.

Gordon M. Bennett and Nancy A. Moran. Heritable symbiosis: The advantages and perils of an evolutionary rabbit hole. *Proceedings of the National Academy of Sciences of the United States of America*, 112(33):10169–10176, 2015. ISSN 1091-6490. doi: 10.1073/pnas.1421388112.

Amrita Bhattacharya, Hannah Tae-Young Pak, and Farrah Bashey. Plastic responses to competition: Does bacteriocin production increase in the presence of nonself competitors? *Ecology and Evolution*, 8(14):6880–6888, 2018. ISSN 20457758. doi: 10.1002/ece3.4203.

Andrew F. G. Bourke. *Principles of Social Evolution*. Oxford Series in Ecology and Evolution. Oxford University Press, Oxford, New York, 2011. ISBN 978-0-19-923115-7.

A. E. Douglas. Mycetocyte Symbiosis in Insects. *Biological Reviews*, 64 (4):409–434, 1989. ISSN 1469-185X. doi: 10.1111/j.1469-185X.1989.tb00682.x.

Angela E. Douglas. Symbiosis as a General Principle in Eukaryotic Evolution. *Cold Spring Harbor Perspectives in Biology*, 6(2), 2014. ISSN 1943-0264. doi: 10.1101/cshperspect.a016113.

Nicole Dubilier, Claudia Bergin, and Christian Lott. Symbiotic diversity in marine animals: The art of harnessing chemosynthesis. *Na-*

ture Reviews. Microbiology, 6(10):725–740, 2008. ISSN 1740-1534. doi: 10.1038/nrmicro1992.

Roberta M. Fisher, Lee M. Henry, Charlie K. Cornwallis, E. Toby Kiers, and Stuart A. West. The evolution of host-symbiont dependence. *Nature Communications*, 8:15973, 2017. ISSN 2041-1723. doi: 10.1038/ncomms15973.

K. R. Foster and T. Wenseleers. A general model for the evolution of mutualisms. *Journal of Evolutionary Biology*, 19(4):1283–1293, 2006. ISSN 1420-9101. doi: 10.1111/j.1420-9101.2005.01073.x.

Steven A. Frank. Genetics of Mutualism: The Evolution of Altruism between Species. *Journal of Theoretical Biology*, 170(4):393–400, 1994a. ISSN 0022-5193. doi: 10.1006/jtbi.1994.1200.

Steven A. Frank. Kin Selection and Virulence in the Evolution of Protocells and Parasites. *Proceedings: Biological Sciences*, 258(1352):153–161, 1994b. ISSN 0962-8452.

Steven A. Frank. Host Control of Symbiont Transmission: The Separation of Symbionts Into Germ and Soma. *The American Naturalist*, 148(6):1113–1124, 1996. ISSN 0003-0147, 1537-5323. doi: 10.1086/285974.

A. Gardner, S. A. West, and A. Buckling. Bacteriocins, spite and virulence. *Proceedings of the Royal Society B: Biological Sciences*, 271(1547):1529–1535, 2004. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2004.2756.

E. Toby Kiers, Robert A. Rousseau, Stuart A. West, and R. Ford Denison. Host sanctions and the legume–rhizobium mutualism. *Nature*, 425(6953):78, 2003. ISSN 1476-4687. doi: 10.1038/nature01931.

E. Toby Kiers, Marie Duhamel, Yugandhar Beesetty, Jerry A. Mensah, Oscar Franken, Erik Verbruggen, Carl R. Fellbaum, George A. Kowalchuk, Miranda M. Hart, Alberto Bago, Todd M. Palmer, Stuart A. West, Philippe

Vandenkoornhuyse, Jan Jansa, and Heike Bücking. Reciprocal Rewards Stabilize Cooperation in the Mycorrhizal Symbiosis. *Science*, 333(6044): 880–882, 2011. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.1208473.

Judith Korb and Duur K. Aanen. The evolution of uniparental transmission of fungal symbionts in fungus-growing termites (Macrotermitinae). *Behavioral Ecology and Sociobiology*, 53(2):65–71, 2003. ISSN 1432-0762. doi: 10.1007/s00265-002-0559-y.

Asher Leeks, Miguel dos Santos, and Stuart Andrew West. Transmission, relatedness, and the evolution of cooperative symbionts. *J Evol Biol Journal of Evolutionary Biology*, 2019. ISSN 1010-061X. OCLC: 8172431913.

Hadeel Majeed, Osnat Gillor, Benjamin Kerr, and Margaret A Riley. Competitive interactions in Escherichia coli populations: The role of bacteriocins. *The ISME Journal*, 5(1):71–81, 2011. ISSN 1751-7362, 1751-7370. doi: 10.1038/ismej.2010.90.

Despoina A.I. Mavridou, Diego Gonzalez, Wook Kim, Stuart A. West, and Kevin R. Foster. Bacteria Use Collective Behavior to Generate Diverse Combat Strategies. *Current Biology*, 28(3):345–355.e4, 2018. ISSN 09609822. doi: 10.1016/j.cub.2017.12.030.

Michael Poulsen and Jacobus J. Boomsma. Mutualistic Fungi Control Crop Diversity in Fungus-Growing Ants. *Science*, 307(5710):741–744, 2005. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.1106688.

Joel L Sachs and Thomas P Wilcox. A shift to parasitism in the jellyfish symbiont Symbiodinium microadriaticum. *Proceedings of the Royal Society B: Biological Sciences*, 273(1585):425–429, 2006. ISSN 0962-8452. doi: 10.1098/rspb.2005.3346.

J. Maynard Smith and G. R. Price. The Logic of Animal Conflict. *Nature*, 246(5427):15–18, 1973. ISSN 1476-4687. doi: 10.1038/246015a0.

Peter D. Taylor and Steven A. Frank. How to Make a Kin Selection Model. *Journal of Theoretical Biology*, 180(1):27–37, 1996. ISSN 0022-5193. doi: 10.1006/jtbi.1996.0075.

Daniel J Thornhill, Kevin T Fielman, Scott R Santos, and Kenneth M Halanych. Siboglinid-bacteria endosymbiosis. *Communicative & Integrative Biology*, 1(2):163–166, 2008. ISSN 1942-0889.

S. A. West, E. Toby Kiers, I. Pen, and R. F. Denison. Sanctions and mutualism stability: When should less beneficial mutualists be tolerated? *Journal of Evolutionary Biology*, 15(5):830–837, 2002a. ISSN 1420-9101. doi: 10.1046/j.1420-9101.2002.00441.x.

Stuart A West, E. Toby Kiers, Ellen L Simms, and R. Ford Denison. Sanctions and mutualism stability: Why do rhizobia fix nitrogen? *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 269(1492):685–694, 2002b. ISSN 1471-2954. doi: 10.1098/rspb.2001.1878.

G. a. K. Wyatt, S. A. West, and A. Gardner. Can natural selection favour altruism between species? *Journal of Evolutionary Biology*, 26(9):1854–1865, 2013. ISSN 1420-9101. doi: 10.1111/jeb.12195.

Joshua Zukewich, Venu Kurella, Michael Doebeli, and Christoph Hauert. Consolidating Birth-Death and Death-Birth Processes in Structured Populations. *PLOS ONE*, 8(1):e54639, 2013. ISSN 1932-6203. doi: 10.1371/journal.pone.0054639.

A Appendix

A.1 Simulations

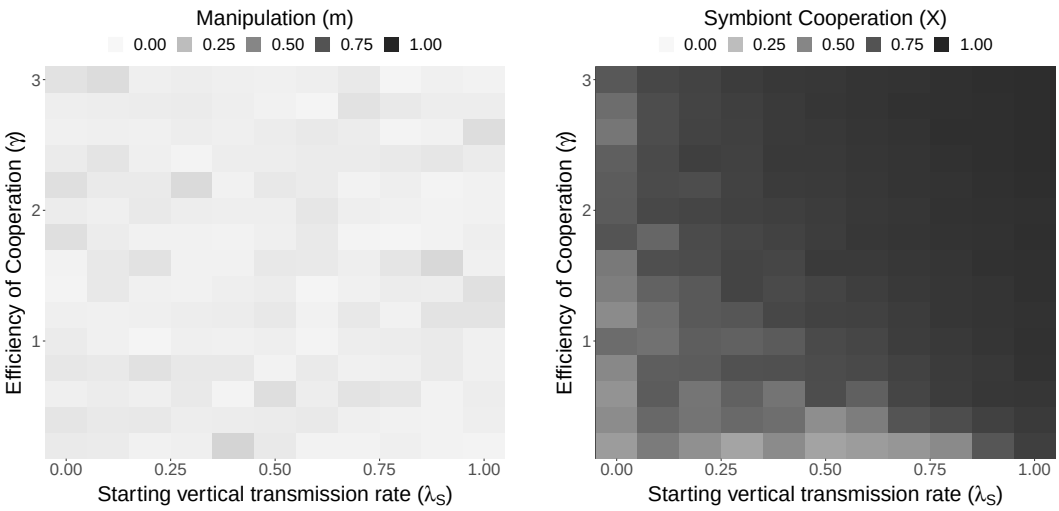
Our individual based simulations are composed of two types of individuals: hosts and symbionts. Each simulation has a fixed number of hosts (N) and a fixed number of symbionts per host (k). Each generation there are N individual reproduction events where one member of the host population copies itself, with a probability proportional to its fitness, and another member dies at random. The order win which tis takes place is randomised to avoid biasing the model to the evolution of certain types of traits (Zukewich et al., 2013).

The fitnesses of the newly created symbionts and the new host are updated using the equations for fitnesses in the main text (eqs. (7) and (8)). Then a new host is chosen and the cycle repeats. By performing a host replacement N times we simulate one generation. Simulations were run for 10000 generations, for the first 2500 generations the host's manipulations trait was not allowed to evolve to allow the symbionts to reach an equilibrium.

The simulations used in this paper were done using a population size of $N = 200$ hosts and $k = 5$ symbionts per host. The cost for the manipulation was kept low but significant at $c = 0.1$. Each simulation was replicated 5 times.

Simulation code was written in the julia programming language version 1.1 and run locally on a Dell Optiplex 7040 computer running Ubuntu 17.10. The starting vertical transmission (λ_S) was sampled in the interval $[0, 1]$ in 0.1 increments. The shape parameter for cooperation (α) was sampled in the interval $[0.2, 3]$ in 0.2 increments.

This sampling gave 165 different combinations which were each simulated over 5 repeats, leading to 825 observations.

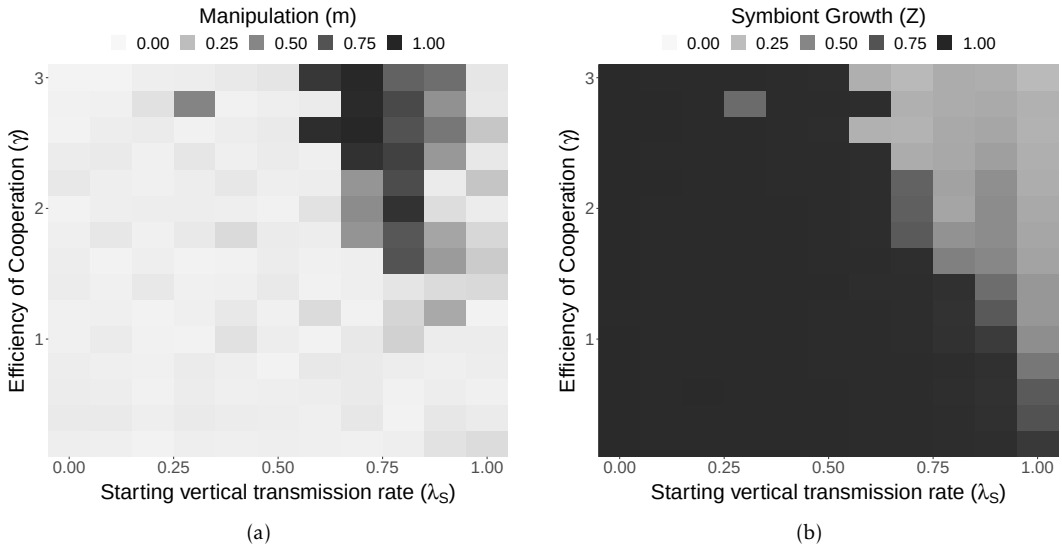


(a) (b)

Figure A.1: **(a)** When the evolution of growth is fixed, $Z = 0.5$, we find that host control doesn't evolve in the modified model with feedbacks. **(b)** When holding growth at a constant ($Z = 0.5$) cooperation in the symbionts evolves to a high level despite low host control. This again leads to no selection for the host to increase control.

A.1.1 Conditional expression plots

We find grossly the same behaviour in the second simulation as in the first. When growth is held constant ($Z = 0.5$) and cooperation (X) is allowed to evolve then investment into bacteriocins is minimal and so host control has no selective advantage (fig. A.1). When instead cooperation is held constant and growth is allowed to evolve ($X = 0.5$) then we find that growth evolves to be lower at high rates of vertical transmission and when cooperation is efficient this leads to host manipulation evolving to reduce the associated increase in bacteriocin production (fig. A.2). Finally when both are fixed we again see no host control evolving. Which is due to manipulation of relatedness not leading to a separation of symbiont pools and therefore benefits cannot be restricted to manipulating hosts which leads to host manipulation being unstable (fig. A.3).



(a) (b)

Figure A.2: **(a)**Similar to what was found in the first simulation with no feedbacks when the evolution of cooperation is fixed, $X = 0.5$, we find that host control can evolve. **(b)** When holding cooperation at a constant ($X = 0.5$) selection for maximal growth ($Z = 1$) seem to relax at high vertical transmission rates (λ_s) and high efficiencies (γ). This selects for host mutants that reduce the conflict generated in those regions.

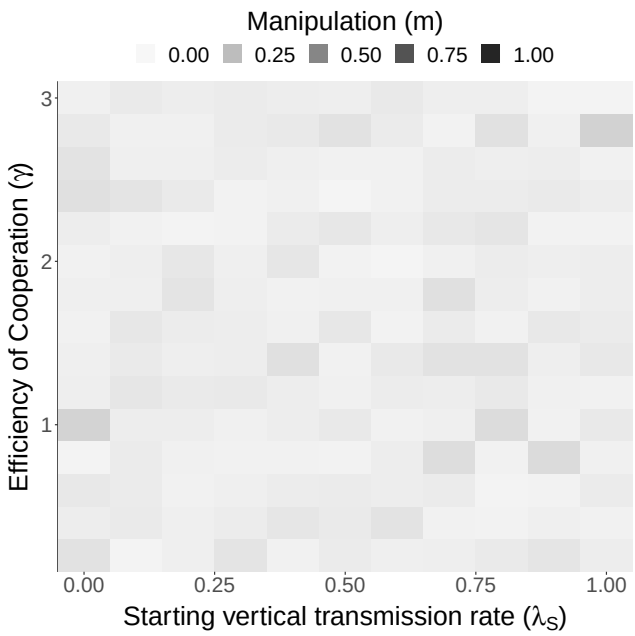


Figure A.3: For the feedback model we again see that when growth and cooperation are fixed, $Z = 0.5$ and $X = 0.5$, host control does not evolve in the parameters explored.